# Dylan Cantor

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## Education

**New York University** Expected May 20
*B.M. Music Technology (Steinhardt); Second Major, Computer Science (CAS); Minor, Mathematics* *GPA 3.*

**NYU Paris — IRCAM program** Spring 2026
*Spatial audio at IRCAM, one day weekly: ambisonics, directivity, binaural rendering* *Paris, France*

**Relevant coursework.** Digital Signal Processing; Machine Learning; Numerical Computing; Artificial Intelligence; Algorithms; Linear Algebra; Probability & Statistics; Analog & Digital Electronics with labs; Analog & Digital Recording; Electronic & Computer Music Synthesis; MIDI Technology II (Max/MSP).

## Research Experience

**Research Team Member — Georgia Tech Center for Music Technology** May 2026 – Present
*Prof. Gil Weinberg's research group* *Atlanta, GA (remote)*

- Sole engineer on the iOS client for **HearMe**, a shipped application that generates a full arrangement from a musician's own playing, returning separable stems, chords, and time-aligned lyrics.
- Built a Web-Audio-style playback engine in React Native with **sample-accurate scheduling** for the piano and drum interfaces, and **offline MIDI rendering** of generated results.
- Implemented the full input path — recording, upload, keyboard capture, lyric entry — into the generation service, with a background watcher that survives app suspension.
- Built the result surface: one transport synchronized across stems, chords, lyrics, and play-along mode, with per-project version history. 200+ commits. Joined for a 150-hour summer term and have remained on the team since, meeting weekly with the group and separately with Prof. Weinberg.

## Professional Experience

**Founder and Sole Engineer — Clubs** February 2026 – Present
*Creative-collaboration platform, live on the iOS App Store* *Brooklyn, NY*

- Built and shipped a platform where projects are posted as cards, discovered by swiping, and applied to; accepted collaborators work in real-time channels. **1,908 users**, 968 of them in the last 30 days.
- Supports **304 active public projects** from 206 creators across 48 schools, with 24,172 swipes, 3,530 applications and 2,133 messages in the last 30 days.
- Designed the AI pipelines: free-text self-descriptions parsed into interests and skills, content moderation, automatic project tagging, semantic search, and an email path that publishes a forwarded message as a card.
- Built a personalized discovery feed on a sawtooth-staircase ranking algorithm, and hardened the SMS one-time-passcode flow against abuse after a live **toll-fraud attack** on the verification endpoint.
- Sole engineer across a multi-package monorepo; 2,300+ commits. Co-founded with Nick Benak, who leads marketing and design.

## Projects

**Jamarama** — real-time multiplayer jam system June 2026

- Led the audio architecture for a system where two to four players each control one dimension of a single fused track from their phones, and one player's chord change re-roots the whole band on the next downbeat.
- Placed the **master clock** in the host browser, fanning beats over WebSocket to phones acting as thin controllers; the lead keyboard fires a **local tone in under 20 ms** so playing never waits on the network.
- Measured the generative model for tempo stability, found it **drifts within roughly 30 seconds** with no tempo input, and removed it from the clock entirely — a deterministic clock-locked band wrapped in an arhythmic texture layer that floats off the grid, where drift is inaudible.
- Ran that layer as **Magenta RealTime 2 on-device via MLX at ∼2.37× real-time**, conditioned on blended player taste, live chord, and crowd energy. *Tone.js, WebSocket, Node, Python, MLX.*

**"I Feel Like"** — second-order ambisonic production and binaural mix Spring 2026

- Produced a multitrack piece entirely in the **ambisonic domain** at IRCAM: roughly twenty sources individually encoded into a second-order field, each with its own room encoding and directivity shaping.
- Applied reverberation inside the ambisonic domain via a feedback-delay-network reverb rather than on a stereo bus, then decoded the same scene two ways — loudspeaker array by all-round ambisonic decoding, headphones by **binaural decoding**.
- Recorded all source material: field recordings on omnidirectional capsules, instruments from the **Baschet brothers' Structures Sonores** sound sculptures, and tetrahedral-microphone material converted A-format to B-format before encoding. *Reaper, IEM Plug-in Suite, Sennheiser AMBEO.*

## Awards

**Winner, Audiotool Challenge** — Google DeepMind Magenta Hackathon June 2026

## Creative and Performance Work

**Cooking Pot** — multi-persona production practice 2022 – Present

- Releases on Spotify and Bandcamp as Finch × Fritz Rango (*rat's nest*, 2024), Ozstriker, Bruner × Whippet, Lord Whippet, and iLLER BOYS — rap, avant-garde jazz-rock, and electronic music. **Close Quarters**, band with Nick Benak. Guitar and modular synthesizer; production and mixing.

## Technical Skills

**Languages and frameworks:** Python, TypeScript, Swift, C++; React Native, Expo, Node, Supabase, Tone.js

**Audio and DSP:** Max/MSP, RNBO, Web Audio API, JUCE, ambisonics and binaural rendering, real-time and streaming audio, sample-accurate scheduling

**Machine learning:** generative audio, music information retrieval, on-device inference (MLX)

**Audio tools:** Reaper, Ableton Live, Logic Pro, IEM Plug-in Suite, VCV Rack